

25.2 A 1.2V 8Gb 8-Channel 128GB/s High-Bandwidth Memory (HBM) Stacked DRAM with Effective Microbump I/O Test Methods Using 29nm Process and TSV

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Increasing demand for higher-bandwidth DRAM drive TSV technology development. With the capacity of fine-pitch wide I/O [1], DRAM can be directly integrated on the interposer or host chip and communicate with the memory controller. However, there are many limitations, such as reliability and testability, in developing the technology. It is advantageous to adopt a logic-interface chip between the interposer and stacked-DRAM with thousands of TSV. The logic interface chip in the base level of high-bandwidth memory (HBM) decreases the C_{IO} , repairs the chip-to-chip connection failure, and supports better testability and improves reliability.

The fundamental structure of HBM is composed of 4-Hi core DRAM and base logic die at the bottom, as depicted in Fig. 25.2.1(a). The core DRAM consists of 2 channels, where each channel has 1Gb density with 128 I/Os and 8 independent banks. Core DRAM has 2n-prefetch with minimum access granularity of 32 bytes [2]. Each channel of core DRAM die has independent address and data TSV with point-to-point (P2P) connection, isolating the operation of every channel. The power and ground of each channel are not isolated and has common plane. The layout in Figure 25.2.1(b) illustrates base logic die architecture. PHY located at the top of the die is for the interface between DRAM and memory controller. PHY has total of 8 channels where a channel consists of an AWORD (address buffer) and four channel-interleaved DWORD (data buffer). The center area is allocated for TSV that deliver signals and power to 4-Hi core DRAM. Similar to PHY, TSV has DDR address and data interface between base and core DRAM. The area between PHY and TSV is filled with signal lines for 1024 I/Os and decoupling capacitance. Some microbumps are depopulated for probe test at chip-on-wafer level. A direct-access (DA) port at the bottom is interface between the automatic test equipment (ATE) and stacked-DRAM. DA port is mainly for testing stacked-DRAM component circuits including cells.

HBM uses semi-independent row- and column-command interfaces, which allow RAS and CAS commands in parallel. A DDR command interface is adopted as well, for reducing pin count. The simplified row- and column-address decoder is shown in Fig. 25.2.2. R[0:5] pins for row addresses and C[0:7] for column addresses aligned by clock can be entered at the same time. In row active command case, there are 4b decoding signals, which are aligned to 4 phase pulse, and other row command (such as precharge, refresh) need 2b decoding signals aligned to 2 phase signal. The column-command operation uses a similar method to the 2b row command. Another function added in HBM is the single-bank refresh. In previous DRAM, the bandwidth efficiency is reduced by all-bank refresh period because DRAM cells cannot be accessed and the bus must idle. In the single-bank refresh function, the refresh cycle can be controlled per unit bank since every bank has a refresh counter and digital t_{RAS} (row active to precharge time) counter. Therefore, any bank that is not in refresh is accessible. HBM uses an unterminated interface, in which bus-switching power is dominant rather than dc power. To reduce this bus switching power, data-bus inversion (DBI-AC) is used.

The primary purpose of base die is the interface with controller and core DRAM. HBM communicates with memory controller through the interposer, which has 1024 I/Os. The interposer has large R and C, causing high power consumption and ISI. The usage of base die reduces C_{IO} significantly and the reduced distance of PHY between DRAM and memory controller decreases the total interface power. The PHY interface circuit is shown in Fig. 25.2.3. Differential clocks (CK_t, CK_c) and 15 DDR addresses are used in AWORD. Per-channel address-parity calculation result is returned to the controller via AERR pin for diagnostic purposes. Differential strobe pairs (WDQS_t, WDQS_c for write, RDQS_t,

RDQS_c for read) are used for a DWORD. There is no skew-compensation circuit, such as DLL, between clock and data. Uncompensated clock strobe aligns data for read output, which saves standby power. There is also a per-DWORD data-parity function for both write and read. The calculated result is returned to the controller via DERR pin. The secondary purpose of base die is to support various test functions. External loopback function between memory and controller is designed for I/O link test and training after assembly. There are four states of loopback register function: resetting register, LFSR mode, register mode and MISR mode [3]. The figure shows a 4b polynomial composed of four flip-flops, XOR gates and feedback signal. In DWORD, there is 20b polynomial ($f(x) = X^{20} + X^{17} + 1$) and there is a 30b polynomial ($f(x) = X^{30} + X^6 + X^4 + X + 1$) in AWORD. HBM DRAM test feature also supports IEEE 1500 protocol interface, which is located in MIDSTACK in PHY region. Using IEEE 1500 protocol, DRAM can support basic connection test functions like boundary scan, chip reset, and internal VREF control, and also complex functions such as loopback, hard repair of DRAM cell after assembly, TSV scan, and memory built-in-self-test.

One of the issues in testing stacked chip is self-verification of the microbump interface. The microbump array has a fine pitch with wide I/O making it difficult to measure the full I/O characteristic directly. Impedance calibration is generally used to reduce PVT variation and this chip has internal resistor calibration, modified by register or fuse. However, it is difficult to measure the impedance of each microbump driver using the conventional method. An impedance-monitoring method is developed for measuring each microbump output drivability and integrity. Each microbump has a boundary scan cell to test and identify pin-to-pin connectivity after assembly. By monitoring enable signal and boundary scan function, one pullup/pulldown driver (of 1408) is enabled. By comparing the average difference of V_{DDQ} current between the case of all drivers off versus each driver on, every driver's drivability and integrity can be accurately measured. Test results of 1408 pins are shown in Fig 25.2.4. Using boundary scan cell and receiver, referenced to a programmable V_{REF} , the V_M level (mid level of pullup and pulldown) is also detected and stored in registers.

There are two ways to test DRAM write/read using DA port. The test circuit and brief flow chart are depicted in Fig. 25.2.5. The first way is pad-loopback, which is parallel write and read operation via microbump using DA port. In DA write case, DA addresses are transmitted through test driver. Simultaneously, DA write data are transmitted through TX and test register. Data are driven onto microbump, which is the same input condition as after assembly. The following write and read operations to DRAM cell are similar to conventional DRAM. In DA read case, core cell data is driven to read register, TX and microbump. To judge data pass/fail in DA mode, output data of read register is transmitted to comparator at the same time. The comparator detects and sends to DA the difference between the written data and read-out data. The second way to test DRAM is using both DA port and serial port. Using serial port WSI, various data pattern (different per DQ) are initialized. Writing these data to DRAM cell is through TX, RX and write register. Read data from DRAM cell goes through read register, TX and RX. The major usage of this method is to check the interface, especially the simultaneous operation of read and TX. Loopback MISR register value can be calculated after multiple read operations. Register value can be transmitted through serial output port WSO, and compared with the expected data. The test result of stacked-DRAM in Fig. 25.2.6 shows 8-channel read operation with 128GB/s at 1.2V and 111GB/s at 1.05V.

There are many design and test challenges in developing stacked-DRAM with microbump interface. System performance is higher using semi-independent row- and column-command interface and single-bank refresh. Parity, loopback and various IEEE 1500 functions support testing between memory and controller. With our methods such as pad-loopback test using DA and impedance-monitoring method, many problems of stacked-DRAM can be observed and distinguished. Four stacked 2Gb DRAM and base die are implemented using 29nm DRAM process. The chip micrograph of core die including TSV structure is shown in Fig. 25.2.7.

References:

- [1] J. Kim, C. S. Oh, et al., "A 1.2V 12.8GB/s 2Gb Mobile Wide-I/O DRAM with 4x128 I/Os Using TSV-Based Stacking," *ISSCC Dig. Tech. Papers*, pp. 496-497, Feb. 2011.
- [2] JEDEC Standard High Bandwidth Memory(HBM) DRAM Specification, 2013.
- [3] B. Keller, T. Bartenstein, "Use of MISRs for Compression and Diagnostics," *Proc. International Test Conference*, pp. 735-743, 2005.

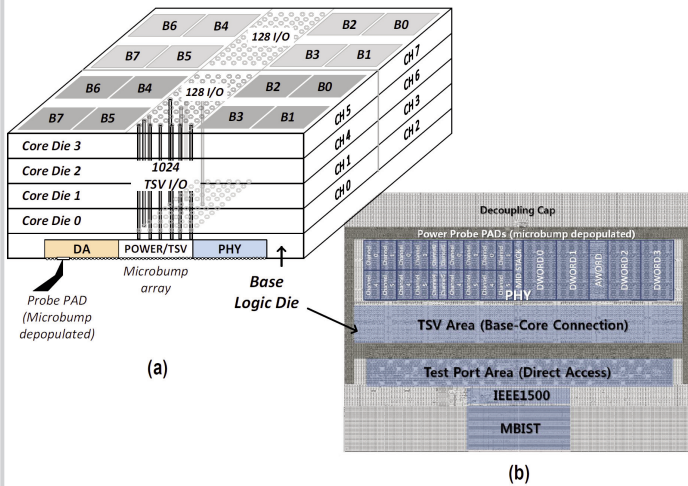


Figure 25.2.1: (a) HBM stacked-DRAM architecture, and (b) layout and functionality of base logic die.

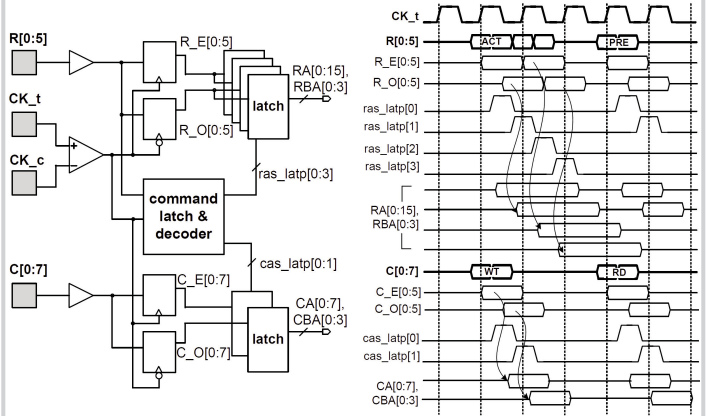


Figure 25.2.2: Semi-independent row/column command interface circuit and timing.

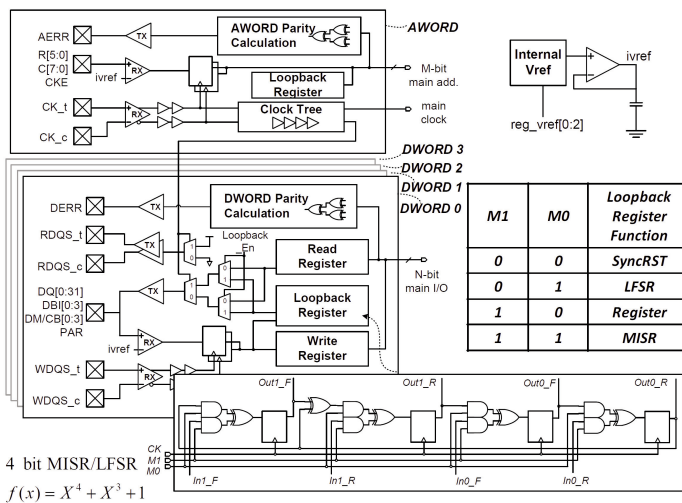


Figure 25.2.3: Clocking scheme of HBM including parity and external loopback circuit.

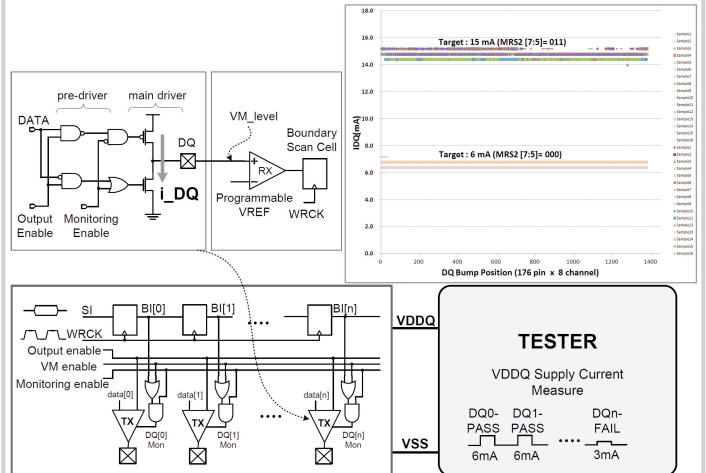


Figure 25.2.4: Microbump impedance-monitoring method and test results.

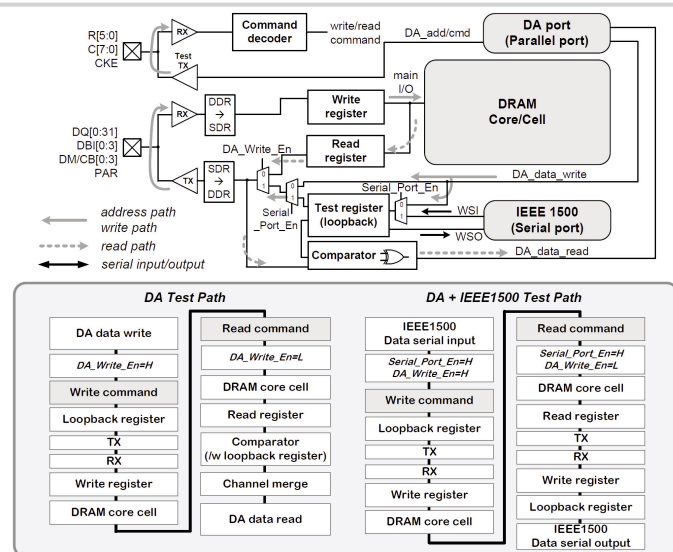


Figure 25.2.5: DA port and serial port test circuit for stacked-DRAM.

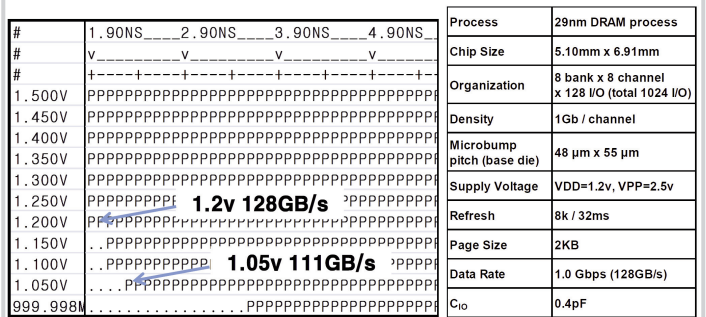


Figure 25.2.6: Stacked-DRAM test results (shmoo plot, 8-channel read operation, 25°C).

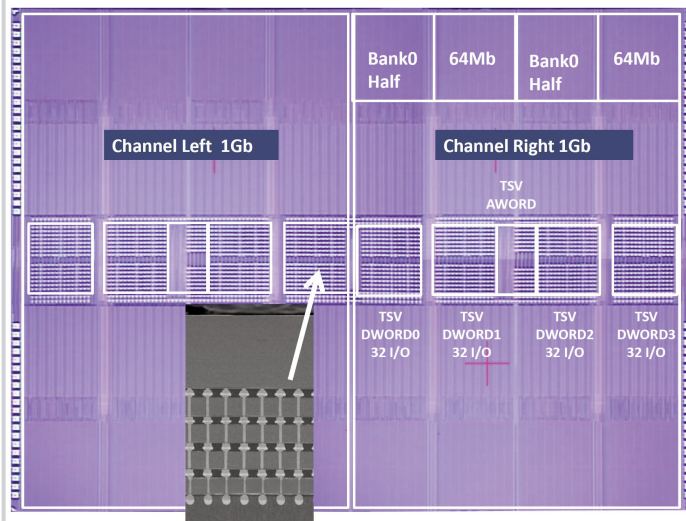


Figure 25.2.7: Chip micrograph including cross-sectional view of TSV.